SANJAY PADMARAJ

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EDUCATION

New Horizon College of Engineering, Bangalore — Expected 2029
B.E. Computer Science & Engineering — CGPA: 9.75 / 10

PROJECTS

CampusOS — unified campus-life platform — github.com/123sanjaypadmaraj/CampusOS
React 18, TypeScript, Vite, Tailwind CSS, TanStack Query, Supabase (PostgreSQL, Auth, Storage, Edge Functions), Playwright
- Built a full-stack web platform that brings campus events, clubs, a community feed, hackathon team-matching, resource booking and issue reporting into one app.
- Designed the backend on Supabase and integrated Razorpay payments (test mode).
- Wrote unit and end-to-end tests (Jest, Testing Library, Playwright) and set up a GitHub Actions CI pipeline; 128 commits.

Club Hub — co-curricular club management app — github.com/123sanjaypadmaraj/Club-App
Next.js, TypeScript, Supabase (PostgreSQL, Auth, RLS), Vitest
- Built one app for ~20 clubs: event registration with waitlist and QR tickets, check-in, feedback and an admin performance dashboard; participant data secured with Postgres row-level security.

AI WordPress Builder — chat-driven website builder — github.com/123sanjaypadmaraj/AI-Wordpress-Website-
TypeScript, Docker, WP-CLI, Playwright, LLMs
- Built an agent that turns a chat description into a real WordPress site in an isolated Docker stack (theme, pages, plugins), with chat edits and undo, audit logging and automated smoke tests.

Aero Engine Digital Twin — Smart India Hackathon prototype — github.com/123sanjaypadmaraj/Aero-piston-engine-simulation
Node.js, Express, Socket.IO, JavaScript
- Simulated a 3-UAV fleet streaming 13 sensors per engine with 9 injectable fault scenarios; z-score and threshold rules predict a health score and remaining useful life on a live dashboard.

Smart Bandage — wearable biosensor software platform — github.com/123sanjaypadmaraj/Smart-Bandage
Python, FastAPI, React, React Native, Docker
- Built the software stack against a simulated sensor: signal processing, FastAPI backend, React and mobile apps, anomaly detection and AI insights (159 tests).

C Visualizer — interactive C learning tool — github.com/123sanjaypadmaraj/Code-Visualizer
Next.js, Tailwind CSS, CodeMirror
- Wrote an in-browser C interpreter that draws arrays, pointers, structs, stack and heap step by step with plain-English explanations; AI features are optional.

Smart Blind Stick — AI + IoT assistive device — **Provisional patent filed**
Sensors, embedded systems, mobile apps
- Designed an assistive stick for visually impaired users combining sensor-based obstacle detection with navigation assistance on embedded hardware.
- Built dual mobile applications, one for the user and one for the caregiver, for real-time monitoring and communication.

NHCE STEM Club Website & RAG Chatbot — newhorizonindia.edu/co-curricular/stem
WordPress, Retrieval-Augmented Generation, Hugging Face LLMs, prompt engineering
- Designed and built the official responsive STEM Club website in WordPress and maintain it with the department.
- Built a RAG chatbot on Hugging Face LLMs, tuned prompts to improve answer accuracy, and deployed it on the live site for student queries.

LinkedIn Profile Matcher — github.com/123sanjaypadmaraj/LinkedIn-Profile-Matcher
Python, Streamlit, Sentence-Transformers, scikit-learn, Ollama
- Built a Streamlit app that compares two LinkedIn profiles (PDF, TXT or JSON): extracts skills and scores similarity using sentence-embedding cosine similarity.
- Added skill-overlap charts and keyword word clouds, plus optional local LLM insights (Mistral, LLaMA, Gemma via Ollama).

AI Data Assistant — github.com/123sanjaypadmaraj/AI_Data_Assistant
Python, Flask, Pandas, Ollama (Mistral)
- Built a Flask app where users upload CSV or Excel datasets and ask questions in plain English; a local Mistral model generates the pandas code that answers each question.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, C, HTML/CSS
Web & Backend:	React, Next.js, Vite, Tailwind CSS, Flask, FastAPI, Node.js, Supabase, PostgreSQL
AI / ML:	RAG, LLM applications, prompt engineering, Hugging Face, Sentence-Transformers, Ollama, scikit-learn, Pandas, NumPy
Tools & Testing:	Git, GitHub Actions, Docker, Jest, Vitest, Playwright, Streamlit
Hardware:	IoT systems, sensor integration, embedded prototyping

ACHIEVEMENTS

- **1st Place:** UDAAN Intra-Institutional Business Plan Competition 2026; E=mc² STEM Expo 2026; Algo Relay (NHCE MCA Dept.)
- **2nd Place:** QuantumX Intercollege Hardware Tech Expo 2026. **3rd Place:** Nexgen Convergence Hardware Hustle Tech Expo
- **Provisional Patent Holder:** Smart Blind Stick for Visually Impaired Individuals

LEADERSHIP & ACTIVITIES

- Active member, STEM Club, NHCE (2025–2026)
- Host, International IEEE Conference ICEMCSI (June 2026)
- Volunteer, BBMP Census of India 2027

CERTIFICATIONS

- Data Analytics (IITM) · AI in Physics (IITM) · C Programming (IIT Bombay)